

Manuel Arnese

Industrial Engineering & Operations Research, Columbia University
306 Mudd, 500 West, 120 street, New York, NY 10027
ma4339@columbia.edu • <https://manuelarnese.github.io/>

RESEARCH INTERESTS

Probability theory; stochastic analysis; optimal transport. Interacting particle systems and propagation of chaos.

EDUCATION

2022 - May 2027 (expected)	PhD, Columbia University. Advisor: Daniel Lacker.
2019 - 2022	MSc, Bocconi University.
2016 -2019	BSc, Bocconi University.

PREPRINTS AND PUBLICATIONS

1. **Manuel Arnese** and Kevin Hu. “Quantitative propagation of chaos and universality for asymmetric Langevin spin glass dynamics.” ArXiv preprint. (2026).
2. **Manuel Arnese** and Daniel Lacker. “Sharp propagation of chaos for mean field Langevin dynamics, control, and games.” ArXiv preprint. (2026).
3. **Manuel Arnese**. “Regularity and propagation of chaos for conditional McKean–Vlasov equations.” *Electronic Journal of Probability* **31** (2026), paper no. 71, 1–40.
4. **Manuel Arnese** and Daniel Lacker. “Convergence of Coordinate Ascent Variational Inference for log-concave measures via optimal transport.” *The Annals of Applied Probability* **36**(3) (2026), 2529–2558.

AWARDS

2022	Unicredit Marco Fanno PhD Scholarship (65,000\$)
2019	Bocconi Master ISU Scholarship
2019	Bocconi Merit Award
2016	Bocconi undergraduate ISU scholarship

INVITED TALKS

1. Probability Seminar, Carnegie Mellon University, Pittsburgh (October 2026).
2. Probability Seminar, University of Washington, Seattle (October 2026).
3. Probability Seminar, University of Michigan, Ann Arbor (September 2026).

4. LiBERA Midterm Meeting, Universidad de los Andes, Bogotá (July 2026).
5. SIAM Annual Meeting, Cleveland (July 2026).
6. 45th Conference on Stochastic Processes and their Applications, Cornell University, Ithaca (June 2026).
7. Probability Seminar, Università di Padova, Padua (April 2026).
8. SIAM Conference on Financial Mathematics and Engineering, Miami (July 2025).
9. ORFE Student Seminar, Princeton University, Princeton (November 2024).

CONTRIBUTED TALKS AND POSTERS

1. Optimal transport: stochastics, projections, and applications, Fields Institute, Toronto (November 2025).
2. Eastern Conference in Mathematical Finance, Carnegie Mellon University, Pittsburgh (October 2025). Poster and lightning talk.
3. InDaM Meeting on Irregular Stochastic Analysis, Cortona (June 2025).
4. SDS Advances in Sampling, Yale University, New Haven (November 2024).
5. ORIE Young Researchers Workshop, Cornell University, Ithaca (October 2024).

TEACHING

Spring 2023 Teaching assistant, Probability for engineers; held recitations and office hours, designed problem sets.

PROFESSIONAL ACTIVITIES

Referee for SIAM Journal of Optimization, Statistics & Probability letters, Biometrika, Stochastic Processes and their Applications.